

Vocabulary

Thinking

Intuition

Gravitation

The force by which objects with mass attract each other.

Universal attractive interaction between all masses, acting along the line joining the center of mass.

Gravitational Force

The pull two masses exert on each other.

A vector force given by $\vec{F}_{12} = G \frac{m_1 m_2}{r^2} \hat{r}$
force $\rightarrow$ always attractive, proportional product of their masses and inversely proportional to the square of their separation

Newton's Law of Universal Gravitation

A rule discovered by Newton that tells us how strongly objects attract each other.

Inertia

How hard it is to change the motion of an object.

Point-like Mass

An object is treated as if all of its mass is concentrated at one point.

Extended Object

An object with size and shape, like the earth.

A mass distribution occupying finite volume, requiring integration or symmetry arguments to compute gravitational effects.

Shell Theorem

A rule that lets us treat spherical objects like point masses.

Inverse Square Law

Gravity gets weaker quickly as distance increases.

A force law where magnitude scales as $1/r^2$.

Vector Force

A force that has both size and direction.

Magnitude of Force

How strong the force is.

Superposition Principle

Multiple gravitational pulls add together (Net gravitational force).

An Introduction to Gravitation

- ↳ Gravity as a Universal Interaction.
- ↳ Mass as the Source of Gravity.
- ↳ From Everyday Objects to the Universe.

→ Falling down

Why do they fall? What determines how strongly they fall?

Gravity as a Universal Interaction:

"The same force that pulls an apple downward is also responsible for the motions of the Moon, the planets and the stars".

The same physical law applies everywhere, to all material objects, without exception.

Gravity is not something the Earth does to us. It is something all objects with mass do to each other.

Why does the Earth pull objects down?

What property of matter allows it to create gravity?

How does the gravitational influence depend on distance?

How do multiple objects influence each simultaneously?

// An object does not need to be alive, active, or moving to exert gravity. Simply having mass is enough. //

Newton's Law Of Universal Gravitation.

"universal interactions between all masses".

Newton was guided by observations:

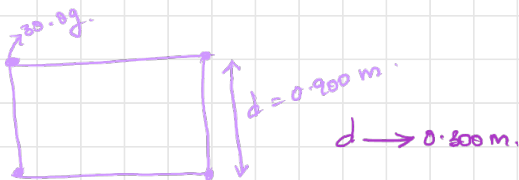
- (i) He knew that heavier objects seem to exert stronger gravitational forces.
- (ii) He knew that gravity weakens distance.
- (iii) He knew forces have both magnitude and direction.

$$F = G \frac{m_1 m_2}{r^2}$$

"Any two point-like masses attract each other with a force that is proportional to the product of their masses and inversely proportional to the square of the distance between them."

8. The figure shows four particles, each of 30.0 g , that form a square with an edge length of $d = 0.900 \text{ m}$. If d is reduced to 0.300 m

Change in the gravitational potential energy of the four-particle system.



Given:

$$m = 30.0 \text{ g} = 0.0300 \text{ kg}$$

$$d_1 = 0.400 \text{ m} \rightarrow \text{Initial side length}$$

$$d_2 = 0.300 \text{ m} \rightarrow \text{Final side length}$$

$$U = - \frac{G m_1 m_2}{r}$$

- Steps:
- (i) How many interacting pairs are there?
 - (ii) Identify all pair distances in the square.
 - (iii) Total gravitational potential energy.
 - (iv) Compute initial and final energies.
 - (v) Change in potential energy.
 - (vi) Plug in the numbers.

For 4 particles:

$$\text{Number of pairs} = \binom{4}{2} = 6.$$

6 $\rightarrow$ pairwise interactions.

Identify all pair distances in the square.

$\rightarrow$ In a square of side length d :

(a) Side pairs.

$\rightarrow$ There are 4 sides

$\rightarrow r_{\text{side}} = d.$

(b) Diagonal pairs.

$\rightarrow$ 2 diagonals

$\rightarrow r_{\text{diag}} = \sqrt{2}d.$



Write the total gravitational potential energy.

$$U(d) = -Gm^2 \int^4$$

Newton's Law of Gravitation

$$\vec{F}_{12} = G \frac{m_1 m_2}{r^2} \hat{r}_{12}$$

unit vector pointing for mass 1 towards mass 2.

Gravity Near Earth's Surface.

General interaction between the masses.

valid anywhere in the universe.

What gravity look like close to the surface of the Earth?

Weight

→ Gravity turns into something we experience.

Weight: What we are really measuring?

weight is not your mass. Weight is a force.

→ Gravitational Force exerted on you by the Earth.

This distinction is important, mass is intrinsic to you, while weight depends on:

- ↳ where you are
- ↳ which planet you are on
- ↳ how massive the planet is

What is the acceleration of particle of mass m due to the gravitational attraction of a particle of mass M ?

Combining Two Fundamental Laws.

Newton's Law of Gravitation

The gravitational force between two masses.

$$F = \frac{GMm}{r^2}$$

→ The force is attractive, directed toward the center of the earth, dependent on both masses and their separation.

Newton's Second Law of Motion.

The acceleration produced by a force

$$a = \frac{F}{m}$$

How objects respond to forces.

$$F = \frac{GMm}{r^2}$$

$$a = \frac{F}{m}$$

Substitute the gravitational force into Newton's second law:

$$a_g = \frac{F}{m} = \frac{GM}{r^2}$$

$$a_g = \frac{GM}{r^2} \rightarrow \text{formula for acceleration when you have an object with mass } m \text{ that is falling.}$$

Insight: The acceleration due to gravity does not depend on the mass of the falling object.

Why All Objects Fall the same way.

All objects fall with the same accelerations in a gravitational field, provided air resistance is negligible.

A feather and a hammer fall together in vacuum.
 Gravity treats them equally in force.

↓
 But because their inertia scales exactly the same way as the gravitational force.

Inertia of the hammer ↑ than the inertia of the feather

Inertia $\propto$ mass $\propto g$.

Define g : The Gravitational Acceleration.

Near the Earth's surface, the distance from the center of the Earth is approximately constant:

$$r \approx R_{\text{Earth}}$$

So gravitational acceleration becomes: $g = \frac{GM_{\text{Earth}}}{R_{\text{Earth}}^2}$